

Experimental Protocol

RNA-seq Differential Expression Analysis Protocol

1. Sample Collection

- Collect tissue samples from treatment and control groups
- Flash-freeze in liquid nitrogen within 30 minutes
- Store at -80C until extraction

2. RNA Extraction

- Use RNeasy Mini Kit (Qiagen)
- Assess quality with Bioanalyzer (RIN > 7.0)
- Measure concentration with NanoDrop

3. Library Preparation

- TruSeq Stranded mRNA Kit (Illumina)
- Input: 1 ug total RNA
- Fragment size: ~200 bp

4. Sequencing

- NovaSeq 6000, 2x150 bp paired-end
- Target: 40M read pairs per sample